FIBRES OF THE MAP ON IRREDUCIBLE COMPONENTS OVER THE ALGEBRAIC CLOSURE ARE EXACTLY THE GALOIS ORBITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the identification of the fibres with the orbits; no step is referred to the reader.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-orbit-irreducible-components is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Irreducible components of a base change and their images in the base. Same chapter, Lemmas lemma-inverse-image-irreducible and lemma-image-irreducible, invoked at proof lines 1777-1780. Supplies that the relevant components map onto the chosen component of X.
- Transitivity of the Galois action on the points above a point of the spectrum of a field. Algebra, Lemma algebra-lemma-Galois-orbit, invoked at proof lines 1781-1784. Supplies that the points above the generic point form a single orbit.
- The Galois action on irreducible components of the base change. Same chapter, Lemma lemma-galois-action-irreducible-components, named at proof lines 1771-1772. Supplies the action whose orbits are being identified with the fibres.

Source reference. Original lemma 04KP: sources/stacks-chow/tags.txt, source lines 5980–5980. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038J: sources/stacks-varieties/pinned-original-source.tex, source lines 1700–1700. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04KX: sources/stacks-varieties/pinned-original-source.tex, source lines 1622–1622. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04KW: sources/stacks-varieties/pinned-original-source.tex, source lines 1596–1596. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field, with separable algebraic closure $\overline{k}$. Let X be a scheme over k. The fibres of the map*

$$\mathit{IrredComp}(X_{\overline{k}}) \longrightarrow \mathit{IrredComp}(X)$$

of Lemma 04KX are exactly the orbits of $\mathit{Gal}(\overline{k}/k)$ under the action of Lemma 038J.

Proof. Let $T \subset X$ be an irreducible component of X. Let $\eta \in T$ be its generic point. By Lemmas 04KW and 04KX the generic points of irreducible components of $\overline{T}$ which map into T map to η. By Algebra, Lemma 04KP the Galois group acts transitively on all of the points of $X_{\overline{k}}$ mapping to η. Hence the lemma follows. □